

Subtract 00:11:00 from timestamps to find place in audio/video recording.

40

00:10:55.810 --> 00:11:03.630

Krzysztof Paruch: Yeah, no worries. Sounds good. Okay, perfect. I'm super excited about the... Presentation today, and

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00:11:03.840 --> 00:11:09.609

Krzysztof Paruch: there's a lot of ground to cover, so let's see what... where we end up. But the last time where we left off.

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00:11:09.960 --> 00:11:16.510

Krzysztof Paruch: was... I think I presented the prototype implementation, I guess, like, that didn't have

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00:11:17.340 --> 00:11:29.210

Krzysztof Paruch: a market logic yet, so, like, the price logic was not implemented yet. And also, we talked a little bit about, kind of, like, the entropy, the information aggregation

44

00:11:29.400 --> 00:11:32.910

Krzysztof Paruch: Potential, that we should look into.

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00:11:33.340 --> 00:11:48.660

Krzysztof Paruch: What I did, is I introduced the price logic into the market. So I took the logarithmic market scoring rule and implemented it. So I basically have a prediction market on-chain that, has this functionality.

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00:11:48.770 --> 00:11:52.600

Krzysztof Paruch: You can actually trade assets at current market prices.

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00:11:52.950 --> 00:11:58.410

Krzysztof Paruch: And, what I found out very, very quickly, which I didn't like at all.

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00:11:58.600 --> 00:12:05.810

Krzysztof Paruch: is that, it doesn't perform as I might have expected it to be. So, like, I mean, like.

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00:12:06.060 --> 00:12:17.439

Krzysztof Paruch: thinking back, it's probably obvious, but intuitively, if you think about it, you wouldn't expect this behavior immediately. So I went down the rabbit hole to explore

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00:12:17.670 --> 00:12:22.830

Krzysztof Paruch: Performance of, different, market...

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00:12:23.970 --> 00:12:31.800

Krzysztof Paruch: implementations or market functions. What do I show you right now? I don't think I show you the thing that I want to show you.

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00:12:32.310 --> 00:12:32.990

Jamsheed Shorish: No, just...

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00:12:32.990 --> 00:12:39.020

Krzysztof Paruch: That's Yeah, I want to show you, actually, this. Okay, now I see that.

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00:12:40.790 --> 00:13:00.560

Krzysztof Paruch: Okay. So this is the implementation that we have, and Danila, you're with us for the first time, so, like, just as a recap, like, I'm building a prediction market on-chain, and this is the prototype that you see on screen. What you can do is you can, like, it's already deployed, but what I'm showing right now is my development environment, which means that the version

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00:13:00.560 --> 00:13:09.719

Krzysztof Paruch: That you see right now is not online available, it's only kind of, like, locally. But there is a lighter version of this online, which does not have the market logic.

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00:13:09.740 --> 00:13:23.959

Krzysztof Paruch: But what you... you can do here immediately with this is, okay, obviously it doesn't log on as quickly as I expected. But, so you can log on with your wallet, so you have a... you can use your credentials to have an account.

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00:13:23.960 --> 00:13:32.940

Krzysztof Paruch: It shows the balance that you have in PTE tokens,

which are the prediction tokens, that you can use for placing bets or buying opinion shares.

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00:13:33.570 --> 00:13:43.940

Krzysztof Paruch: And then you just, participate in those, polls, or those, those markets, and,

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00:13:44.290 --> 00:13:46.520

Krzysztof Paruch: What you have here is...

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00:13:46.730 --> 00:13:57.770

Krzysztof Paruch: Let me log on maybe later. So what you have here is kind of, like, those tabs where you switch between active markets, markets that are pending resolution, and markets that are

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00:13:57.940 --> 00:14:06.619

Krzysztof Paruch: resolved, which I don't have at the moment, but still, like, you can switch back and forth. But you do have some, some that are pending resolution.

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00:14:06.810 --> 00:14:25.109

Krzysztof Paruch: And the ones that are active. And what you see here is, like, your options to participate in the market, so you can buy shares, yes, no. This is the current market standing, so, like, you would see that, kind of, like, 8 people, or, like, basically me, bought 8 shares of OPS, and then 5 shares of no.

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00:14:25.120 --> 00:14:40.189

Krzysztof Paruch: You see the market statistics, and you see, like, how much collateral you have collected in the meantime, and also you see the bounded loss maximum that you have, potentially to suffer. And you do also see the price history. And this is, like, what I wanted to mention in the very beginning.

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00:14:40.230 --> 00:14:51.769

Krzysztof Paruch: what you see immediately when you look at this is that you get different realizations of market prices, obviously. Now I say... now I say obviously, but I didn't say this when I started building this.

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00:14:51.770 --> 00:15:02.240

Krzysztof Paruch: based on both the rule, the scoring rule that you

implement, and its parametrization. So, depending on the function, the trading function that you have.

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00:15:02.380 --> 00:15:09.680

Krzysztof Paruch: And, all the kind of, like, the trading structure, so if it's, like, an order book market, or if it's, like, an automated market maker.

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00:15:09.880 --> 00:15:17.920

Krzysztof Paruch: You would get different implications based on signal inputs, or any kind of, like, event that is happening.

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00:15:18.000 --> 00:15:32.290

Krzysztof Paruch: And you would have different manifestations. And what you see immediately here is, like, a comparison between, like, what would be the simpler ratio between the shares that you have outstanding. So in this context, when you have 8 shares yes and 5 shares no, you would

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00:15:32.370 --> 00:15:35.679

Krzysztof Paruch: Have to have the probability of 61 to 38.

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00:15:35.850 --> 00:16:00.619

Krzysztof Paruch: of this event, realized, realizing, which is also, represented here on the... on the kind of, like, current part of the market prices that you see here. So, and if I do... do trades here, which obviously now in the demonstration do not work, like, for some reasons, maybe I can log on later, but, like, if you... if you acquire shares, like, you would obviously update those, those, those, those, rates.

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00:16:00.620 --> 00:16:03.059

Krzysztof Paruch: Ratios, and also those paths here.

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00:16:03.100 --> 00:16:15.740

Krzysztof Paruch: But what you see immediately happening is that you, kind of like on the market scoring rule chart, where you calculate the prices of shares based on the partial derivatives of those,

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00:16:16.260 --> 00:16:27.280

Krzysztof Paruch: Holdings or inventory changes, from the cost function that you have. You get a completely different, pricing.

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00:16:28.110 --> 00:16:50.970

Krzysztof Paruch: So, it's... it... I mean, this is kind of like the three-dimensional graph of the cost function hyperplane, which is also not super, super, like, intuitive, but, like, it's here, you can... you can look at it. And, you see here, this is the relevant calculation. So, basically, the cost function that you have right now, and, if you take the partial derivative with respect to kind of, like, option A, which is in this case, yes.

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00:16:50.970 --> 00:17:00.939

Krzysztof Paruch: you would have the price of \$50.75, and in the other direction, you have \$49.25, right? So, like, when I implemented this, and when I look at this, I thought.

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00:17:00.940 --> 00:17:10.739

Krzysztof Paruch: okay, this is quite... quite far off from what I had intuitively expected, because I would have thought, at least, that this would be looking something similar to this, right?

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00:17:11.060 --> 00:17:29.869

Krzysztof Paruch: But now I know that it is completely dependent on both, like, selection of market function plus its parametrization, plus it's dependent on, kind of, like, the current market debt, plus it's obviously dependent on, any signals that you feed in, so the delta between those shares, right? So...

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00:17:29.870 --> 00:17:36.720

Krzysztof Paruch: just to sum it up, like, what I found out, and what I'm happy to talk about today, or, like, discuss with you, is, that...

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00:17:37.790 --> 00:17:46.070

Krzysztof Paruch: There is, a local behavior of the cost function AMM,

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00:17:46.870 --> 00:17:55.260

Krzysztof Paruch: That is kind of similar to what you would expect intuitively, but for the edge cases, you get into

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00:17:55.410 --> 00:18:04.489

Krzysztof Paruch: difficult territory. So, those things do not add up or match, similar as you see it here, right? So,

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00:18:04.860 --> 00:18:12.089

Krzysztof Paruch: I guess that my recommendation, based on what I have learned so far, is that in order to set up

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00:18:12.610 --> 00:18:14.150

Krzysztof Paruch: NMM.

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00:18:14.600 --> 00:18:24.139

Krzysztof Paruch: you would need to anticipate market depth, and then parameterize it accordingly. So, like, there are some estimations that you can make based on,

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00:18:24.490 --> 00:18:28.080

Krzysztof Paruch: Number of shares outstanding and the delta between the shares that you have.

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00:18:28.260 --> 00:18:35.099

Krzysztof Paruch: That would inform you how you should select, the parameter B of the logarithmic market scoring rule.

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00:18:35.440 --> 00:18:36.480

Krzysztof Paruch: function.

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00:18:37.300 --> 00:18:50.260

Krzysztof Paruch: And then you would locally, around this neighborhood of kind of, like, market depth, have nice behavior, but, but otherwise, you would, yeah, differ.

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00:18:50.300 --> 00:18:58.880

Krzysztof Paruch: maybe let's... let's stop a little bit. So, like, kind of, like, this is... this is, any feedback or, like, any comments on that, any thoughts here?

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00:19:00.080 --> 00:19:02.910

Jamsheed Shorish: One of the things that comes to mind is.

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00:19:02.940 --> 00:19:09.830

Krzysztof Paruch: At least from the point of view of parameterization, is that you have a lot of degrees of freedom on how you allocate.

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00:19:09.830 --> 00:19:17.070

Jamsheed Shorish: The shares that accrue given a particular price, or given a particular demand, as a function of depth.

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00:19:18.020 --> 00:19:20.949

Jamsheed Shorish: Right. You don't necessarily have to...

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00:19:21.090 --> 00:19:28.369

Jamsheed Shorish: make a prediction on what is going to be expected in the future. Instead, you can work the other way, and only issue

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00:19:28.800 --> 00:19:36.559

Jamsheed Shorish: The kind of amounts of shares for individuals on the basis of a price that keep you on the invariant surface.

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00:19:36.720 --> 00:19:41.140

Jamsheed Shorish: That match up to the prices that you want to have reflective of the probabilities.

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00:19:41.880 --> 00:19:55.939

Jamsheed Shorish: So, what that does is it changes the implicit price. Like, the market ask or the market bid becomes a thing that people then, as they make their purchases, their votes, they buy one way or the other.

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00:19:56.010 --> 00:20:05.939

Jamsheed Shorish: They actually end up swinging the future price in one direction or the other, which keeps you on the indifference surface, as we might call it, or the invariant surface.

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00:20:06.130 --> 00:20:07.299

Jamsheed Shorish: of the system.

100

00:20:07.450 --> 00:20:16.109

Jamsheed Shorish: And that would keep you from having that divergence, or having to do any kind of predictability, because you would basically be skewing the system so that it was always maintaining that invariance.

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00:20:20.250 --> 00:20:24.029

Krzysztof Paruch: I'm not quite sure if I follow, so, like, what does this mean practically?

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00:20:24.030 --> 00:20:24.650

Jamsheed Shorish: Optic.

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00:20:25.110 --> 00:20:27.920

Jamsheed Shorish: Yeah, so, so think in the sense of,

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00:20:27.920 --> 00:20:30.830

Krzysztof Paruch: like, think of a typical Uniswap style.

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00:20:30.940 --> 00:20:35.280

Jamsheed Shorish: AMM. So, essentially, you've got an invariant surface that you're trading along.

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00:20:35.800 --> 00:20:50.419

Jamsheed Shorish: Right. Such that you're always maintaining that particular constant proportionality. Of course, we're abstracting away from, you know, the feed collection and the moving of that surface, but that surface is a surface of constant exchange. There's a constant trade-off that's being performed.

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00:20:50.490 --> 00:21:03.119

Jamsheed Shorish: up to the discreteness, and so the discreteness then gives you some little differences that you need to take into consideration where you can't use a partial differentiation. You have to use, you know, the actual changes, the deltas instead of the D s, so to speak.

108

00:21:03.440 --> 00:21:08.889

Jamsheed Shorish: But you're walking along that indifferent surface, and there aren't trades that exist which are off of that surface.

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00:21:09.580 --> 00:21:26.159

Jamsheed Shorish: Here you have a cost function, a cost functional, that in itself has a set of parameters that you could specify a priori, and then ensure that the kinds of trades at the offers that are made in the future respect that cost functional at all times.

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00:21:27.470 --> 00:21:29.949

Jamsheed Shorish: And what that will do is, it will...

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00:21:30.150 --> 00:21:38.500

Jamsheed Shorish: Tie you down to having certain price ratios that are reflective of the kinds of probabilities that you would calibrate

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00:21:38.710 --> 00:21:50.810

Jamsheed Shorish: according to how the system would be moving. And so, you would basically say, I want the probabilities to be this way, this particular way, or the reflective of this particular initial condition. That would be, like, the starting parameterizations.

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00:21:50.870 --> 00:22:00.570

Jamsheed Shorish: But then the prices themselves would ebb and flow depending upon whether people are overwhelmingly likely to say yes for Valentine's Day versus no, for example.

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00:22:01.230 --> 00:22:02.049

Jamsheed Shorish: So you'd be true.

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00:22:02.950 --> 00:22:03.820

Jamsheed Shorish: surface.

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00:22:04.480 --> 00:22:08.449

Krzysztof Paruch: But are you saying that, you can make

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00:22:08.650 --> 00:22:23.149

Krzysztof Paruch: the cost function behave as if it was, like, a simple ratio of the outstanding shares always, or is your argument still that, like, this works locally, or, like, at least to some extent, and it doesn't work

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00:22:23.620 --> 00:22:24.319

Krzysztof Paruch: In the...

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00:22:24.320 --> 00:22:34.830

Jamsheed Shorish: Well, I don't know. That's... yeah, that's something you'd have to explore to see whether you could actually pin down both degrees of freedom so that you had a reflection of the population probabilities.

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00:22:35.040 --> 00:22:41.909

Jamsheed Shorish: One of the questions that kind of comes up that is still worthy of exploring, though, and I think this is one of the things that maybe is

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00:22:42.260 --> 00:22:53.060

Jamsheed Shorish: Like, even thinking before this step is to ask yourself whether or not you really want the probabilities to reflect simply the relative ratios, that simple ratio.

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00:22:53.150 --> 00:23:04.209

Jamsheed Shorish: Is that the most intuitive method of understanding whether individuals who are buying in are reflecting the actual probabilities of this particular thing materializing.

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00:23:04.210 --> 00:23:21.050

Jamsheed Shorish: Or are you getting things like Keynesian beauty contest-type results, where people are just piling on in a direction without necessarily knowing whether or not the underlying probabilities are respected? Because that, of course, is a behavioral difference that you can't necessarily disentangle from the simple ratio.

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00:23:21.190 --> 00:23:32.430

Jamsheed Shorish: So, I don't want to influence your direction one way or the other. If you think that the simple ratio is the way in which the probabilities must be defined, then we would move to the next question of how do you calibrate the cost function to reflect that.

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00:23:32.560 --> 00:23:44.010

Jamsheed Shorish: But on the other hand, because that's in a dynamical system that you're then trying to find the equilibrium for, and then the marginal prices are going to be the ones that keep you on that indifferent surface at each tick in time.

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00:23:44.240 --> 00:23:57.610

Jamsheed Shorish: But perhaps it's worth reflecting, you know, do you really care about the simple ratio as the statistic or the best possible function that's forecasting forward the probability of an actual occurrence of this.

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00:23:57.610 --> 00:23:58.800

Krzysztof Paruch: Yeah, yeah.

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00:23:59.050 --> 00:24:00.230

Jamsheed Shorish: Yeah, so...

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00:24:00.830 --> 00:24:07.359

Krzysztof Paruch: Yeah, so I haven't thought about, kind of, like, the validity of this assumption, but I basically very quickly just,

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00:24:08.380 --> 00:24:26.260

Krzysztof Paruch: thought, okay, this must be it, right? It's like, I don't know why, like, I didn't question it at all, but for me, like, what I assumed is kind of, like, the, like, very, very right way to go here is to... to have this simple ratio at least close to what my prices are.

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00:24:26.380 --> 00:24:37.690

Krzysztof Paruch: Because otherwise, if I don't have it, like, I lose the... information aggregation, Interpretation, of probabilities here.

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00:24:37.900 --> 00:24:40.600

Krzysztof Paruch: or...

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00:24:40.600 --> 00:24:50.390

Jamsheed Shorish: one-to-one, right? It's a functional form assumption that passes from the information... you're selecting a functional form for information aggregation when you choose that. You could select another functional form.

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00:24:50.710 --> 00:25:06.200

Jamsheed Shorish: for the information aggregation. In other words, probabilities don't rely one for one on the relative ratios of purchases, rather they are a function of dot dot dot. And that way you still have information aggregation occurring, it's just the realized probabilities will be different.

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00:25:07.180 --> 00:25:26.169

Jamsheed Shorish: In a way that gets back to that discussion we had last time about stake prices, and that state pricing, again, I always kind of come back to that. I sort of hang my hat of intuition upon that, that notion that if I were looking at something and then trying to evaluate that from a risk-neutral perspective, I actually get

probabilities that are used to evaluate my decision.

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00:25:26.290 --> 00:25:32.969

Jamsheed Shorish: Here on asset investment, as a risk-neutral investor, that depart from the objective probabilities.

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00:25:33.750 --> 00:25:58.560

Krzysztof Paruch: Okay, so this is cool, because I didn't expect that, and I also didn't thought about that, that I have the degree of freedom to kind of, like, lose this assumption at all. So, like, for me, it was, like, straightforward, like, this has to be it, right? Like, I have to converge to those numbers. Somehow, this is the only thing that tells me... allows me to make any interpretation of probabilities here, but it sounds like it's not the case. Like, it sounds like I can be more flexible with it, I have to understand what's happening, though.

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00:25:58.560 --> 00:26:03.629

Krzysztof Paruch: And, like, how to make the proper interpretation of what I have. But it looks like,

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00:26:03.860 --> 00:26:11.090

Krzysztof Paruch: those numbers that I would expect to be represented by the outstanding shares, direct proportion, And not to, like...

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00:26:11.280 --> 00:26:19.310

Krzysztof Paruch: always one-to-one Tied to the probabilistic... interpretation.

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00:26:20.400 --> 00:26:32.419

Jamsheed Shorish: That's correct. And the reason for that is, in a way, the reason for that is simple in the sense that the questions that are often being asked, that are realizations of an event.

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00:26:32.610 --> 00:26:34.640

Jamsheed Shorish: Are things that themselves

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00:26:34.760 --> 00:26:48.080

Jamsheed Shorish: may have an objective probability, they may be a subjective probability, they may actually be probabilities that are dependent upon people's opinions or perspectives themselves. There's a feedback loop that loops back into that, none of which you can

capture.

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00:26:48.930 --> 00:26:54.939

Jamsheed Shorish: In a simple ratio. That ratio itself doesn't tell you anything about whether or not an objective

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00:26:55.300 --> 00:26:59.430

Jamsheed Shorish: realization is going to occur. And in a way, that's something you don't...

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00:27:00.170 --> 00:27:18.749

Jamsheed Shorish: In a sense, you don't really care... well, I should ask and say it somewhat differently. From a perspective of revenue generation, suppose you were somebody who was trying to run this market for a profit, in a way, you don't care about the objective realization. What you really care about instead are what are the probabilities that will cause different types of trades to occur.

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00:27:18.750 --> 00:27:21.710

Jamsheed Shorish: Different types of buying and selling patterns to occur.

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00:27:21.870 --> 00:27:23.500

Jamsheed Shorish: On the basis of...

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00:27:23.640 --> 00:27:31.400

Jamsheed Shorish: How do people feel that they are going to be getting as a part of what their behavior is, as a part of the whole, relative to

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00:27:31.710 --> 00:27:48.920

Jamsheed Shorish: doing things, let's call it quote-unquote, honestly, and only buying if and only if the objective probability is reflected. Because those are two different regimes, and in a way, as the owner of the marketplace, you really want to have as wide a possible... possibility for yourself to make money

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00:27:48.920 --> 00:28:05.550

Jamsheed Shorish: as you can, which means you're going to be including these subjective probabilities or these risk-neutral probability assessments, and you'll move further and further and further away from the objective probabilities that are occurring. They are, of course,

important because they are setting the aggregation standard by which you are bringing everybody together.

152

00:28:05.820 --> 00:28:16.159

Jamsheed Shorish: But, at the same time, if you were wanting to make money, and you were just hyping something, you could get pretty far away from the objective probabilities and still make a fortune.

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00:28:17.000 --> 00:28:19.440

Krzysztof Paruch: Okay, so this sounds good. Yeah, okay, yeah.

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00:28:20.280 --> 00:28:36.299

Jamsheed Shorish: So, yeah, anyway, just to sum up, that departs from the mantra of, this is a prediction market, and the prediction market is supposed to be reflecting what is actually occurring, because you are now moving in more to how do individuals kind of forecast their own ability to use this as an asset.

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00:28:36.620 --> 00:28:50.690

Jamsheed Shorish: that happens to be tied, but there are still going to be pass-throughs of the information aggregation that people may have, and that's the revelation of what their information is, versus them simply following the masses. The problem is that it's very difficult to disentangle those two.

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00:28:50.830 --> 00:28:51.510

Jamsheed Shorish: So, thank you.

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00:28:52.280 --> 00:29:06.669

Krzysztof Paruch: No, this is very cool, I haven't thought about that. I mean, like, how does this all change if we kind of, like, add another market? Like, have two marketplaces? Because, like, you would expect, some alignment, right, between those two then, so...

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00:29:07.160 --> 00:29:16.949

Jamsheed Shorish: If they're actually substitutable, if they are, in fact, Aero securities, in the sense that they are, in fact, paying 1, if that exact event occurs and zero otherwise.

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00:29:17.120 --> 00:29:34.140

Jamsheed Shorish: then as long as there are no market frictions between them, you would expect them to equalize, because eventually

people would be totally indifferent about which marketplace they wanted to invest in, because they're receiving exactly the same asset, the claim on the asset that they're purchasing. If there are any departures, though, or if there's even a subjectivity.

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00:29:34.140 --> 00:29:40.209

Jamsheed Shorish: over what is being asked for in terms of what state of nature is occurring. Then what you're looking at, like, from the

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00:29:40.340 --> 00:29:53.210

Jamsheed Shorish: If you think about it, like, from the sigma algebra point of view, instead of having a kind of an unambiguous event that is occurring, what you are instead saying is, hey, there's a probability distribution over the subset of things that could be occurring.

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00:29:53.350 --> 00:30:07.490

Jamsheed Shorish: And if two markets actually disassociate according to a partition of that subset, they're going to have different prices for what we might call, instead, similar subset-based event realizations.

163

00:30:10.030 --> 00:30:21.089

Jamsheed Shorish: Right. It's going to depend a lot upon, yeah, on the way in which those events are being articulated. They have to be precise, because aero securities only pay in the event of one specific thing occurring.

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00:30:21.960 --> 00:30:26.079

Krzysztof Paruch: Right, yeah, but in the case of, like, Super Bowl champion, which was kind of like an event.

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00:30:26.510 --> 00:30:32.920

Krzysztof Paruch: Earlier this week, you would basically expect an market equilibrium, right?

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00:30:33.620 --> 00:30:40.740

Jamsheed Shorish: If it were just binary, who wins? But if it were instead, who wins in which quarter was the winning touchdown?

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00:30:40.850 --> 00:30:43.850

Jamsheed Shorish: thrown. Who wins by what spread?

168

00:30:44.260 --> 00:31:02.420

Jamsheed Shorish: who wins, how many overtimes were there? Those are all the things that people actually... in real gambling markets, they bet on all of these facets that are still in the same overarching who wins, who loses, but then there are other qualities or properties that need to obtain before the, so to speak, the arrow security pays off.

169

00:31:03.300 --> 00:31:11.790

Krzysztof Paruch: Alright, yeah, cool. Okay, understood. This is very nice. Thank you so much for this context. I haven't thought about this perspective at all. So, like, I was...

170

00:31:12.420 --> 00:31:30.619

Krzysztof Paruch: like, very forcefully trying to get an understanding of, like, why is it so far away, and, like, how to, like, what is the difference between those two markets? So, basically, the rest of the talk now would be, experiments that I've conducted to investigate performance between those two market families.

171

00:31:30.620 --> 00:31:33.730

Krzysztof Paruch: And kind of, like, also digging deeper.

172

00:31:33.830 --> 00:31:39.900

Krzysztof Paruch: Into different... both cost functions, but also parameterizations of the... a decentralized case.

173

00:31:40.010 --> 00:31:53.620

Krzysztof Paruch: And what I did is I collected, real market data from two markets, so, like, I took the biggest poll in market markets, where you can actually grab data, which is one is

174

00:31:53.720 --> 00:31:56.109

Krzysztof Paruch: will Bitcoin reach...

175

00:31:56.310 --> 00:32:13.740

Krzysztof Paruch: some price, I think 100K to 2024, and the other was actually the Super Bowl champion winner, this week. And, I built pipelines to call data from the graph, from polymarket.

176

00:32:15.000 --> 00:32:20.349

Krzysztof Paruch: And from something else, like, there is a third source that I used as well.

177

00:32:20.550 --> 00:32:32.889

Krzysztof Paruch: But they have some limitations. So, like, the best one is actually the graph or, getting full historic data, but if you want to look into order book style data, you can

178

00:32:33.390 --> 00:32:39.279

Krzysztof Paruch: You can collect it from Poland market directly, but there are some limitations with respect to how far you can go back in history.

179

00:32:39.920 --> 00:32:51.639

Krzysztof Paruch: Yeah, and I've run those experiments just to see how this works. I'm happy to share this now, so let me just grab the notebook that I have.

180

00:32:51.760 --> 00:32:54.750

Krzysztof Paruch: And, show this to you.

181

00:32:56.330 --> 00:32:59.050

Krzysztof Paruch: Which should be here.

182

00:33:00.180 --> 00:33:02.560

Krzysztof Paruch: So this is the one that I have.

183

00:33:03.900 --> 00:33:07.370

Krzysztof Paruch: So let's look first at,

184

00:33:07.700 --> 00:33:21.000

Krzysztof Paruch: artificial data, and this is the market that I pulled, so will Bitcoin hit 100K in 2024? So, there was a lot back and forth between, kind of, like, how to set up the data feed and the pipeline for data collection, so, like.

185

00:33:21.470 --> 00:33:32.039

Krzysztof Paruch: I spent a lot of time on that, but what I'm gonna show you now is really more on the analytical side. What, like, how does it, behave?

186

00:33:32.540 --> 00:33:44.700

Krzysztof Paruch: And what you see here are, the signals that I feed in. So, like, I defined, just the, general signals that I want to investigate.

187

00:33:44.880 --> 00:33:51.830

Krzysztof Paruch: Which is kind of, like, flat signals here, a Gaussian, signal, then, drift.

188

00:33:52.100 --> 00:34:00.729

Krzysztof Paruch: With... with some, saturation, a step... one single step function, a different size and different levels.

189

00:34:01.240 --> 00:34:09.030

Krzysztof Paruch: And then mean reverting signal, and, finally, heavy-tailed...

190

00:34:09.139 --> 00:34:13.730

Krzysztof Paruch: At the bottom. And feeding this into, kind of, like, all of the...

191

00:34:14.080 --> 00:34:25.819

Krzysztof Paruch: different, yeah, parameterizations, so you see what happens to the prices if you have, different, functions, right, and different parameterizations.

192

00:34:25.889 --> 00:34:36.640

Krzysztof Paruch: And I think that, yeah, so this looks exciting, right? So, like, so you can look into this and say, okay, there is completely different behavior,

193

00:34:36.800 --> 00:34:44.319

Krzysztof Paruch: And different impacts, of, like, the price, conditioned on, on the market that you choose.

194

00:34:44.489 --> 00:34:53.500

Krzysztof Paruch: and the parameterization that you choose. And this is where I basically landed with this analysis. So, like, okay, it,

195

00:34:53.639 --> 00:35:07.850

Krzysztof Paruch: It shows me, that, kind of, like, the pattern is still the same, but the amplitude is different, and that you get, like, to, to a rail, or to, like, flattening, like, you over-exceed, right, like, the... the edge cases.

196

00:35:07.970 --> 00:35:12.380

Krzysztof Paruch: So, so... This is then...

197

00:35:13.150 --> 00:35:25.890

Krzysztof Paruch: what I expected, based on, kind of, like, my intuition, right? Like, okay, if you have now different parameterizations, and you feed in the same signal, this is how it would behave, based on... on what you've selected.

198

00:35:26.410 --> 00:35:43.180

Krzysztof Paruch: And then, what I did as well is I tried this with actual data, so I pulled, not kind of like the signals that artificial ones, but the actual signals from polymarket, and the generated signals, and compared the order bookcase to the

199

00:35:43.710 --> 00:35:53.950

Krzysztof Paruch: To the LSMR, LSMR case, which is like this, right? So, like, this is basically the price that you've seen for this market on... on PolyMarket, the green one.

200

00:35:54.180 --> 00:36:00.560

Krzysztof Paruch: And the red one is the one that is simulated by the same interactions

201

00:36:01.410 --> 00:36:09.130

Krzysztof Paruch: In a logarithmic markets, current rule market, with this parameterization. B is 50, and alpha is 0.006 second.

202

00:36:09.570 --> 00:36:25.889

Krzysztof Paruch: And what you see here is obviously the saturation, you get to this much quicker, and then you see some oscillation here. So, like, it completely, yeah, it, like, doesn't perform anymore. So, like, it works completely away from anything that is, informative.

203

00:36:25.940 --> 00:36:31.710

Krzysztof Paruch: In... in this context. Obviously, you can ask if those trades

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00:36:31.880 --> 00:36:36.359

Krzysztof Paruch: would have happened on the LMS error market,

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00:36:36.940 --> 00:36:39.789

Krzysztof Paruch: They would definitely not if they are coupled.

206

00:36:39.970 --> 00:36:47.179

Krzysztof Paruch: And they might not have... if... even if it's isolated, right? Because people would,

207

00:36:47.500 --> 00:36:59.740

Krzysztof Paruch: behave differently. So, like, they would have their expectations, or they would have their intuition about the probability of an event, and they would, have not continued with the same trading behavior, they would have reacted differently.

208

00:37:00.000 --> 00:37:15.900

Krzysztof Paruch: But I guess that the analysis that we have here, as it is, is kind of, like, good enough to answer the questions that I had initially, or, like, to help me build further intuition with respect to, you know, how those two different markets

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00:37:16.970 --> 00:37:18.360

Krzysztof Paruch: evolves.

210

00:37:19.180 --> 00:37:25.179

Krzysztof Paruch: Yeah, maybe let's stop here a little bit and just discuss if there is some... if there are some comments.

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00:37:39.430 --> 00:37:44.159

Krzysztof Paruch: Okay, if not, then, basically, what,

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00:37:45.590 --> 00:37:56.600

Krzysztof Paruch: What is left to do here is to do actual analysis on this data, and to investigate, kind of like, those patterns, this behavior, so just to see

213

00:37:56.730 --> 00:37:58.770

Krzysztof Paruch: What...

214

00:37:59.680 --> 00:38:11.449

Krzysztof Paruch: yeah, what do... what sense to make out of this behavior? Like, it supports the intuition, yes, okay, there's different behavior and there is, like, something happening, but the first task here would be to make a proper interpretation.

215

00:38:11.810 --> 00:38:21.929

Krzysztof Paruch: The second would be to, kind of, like, ask the questions that I asked before. What, like, is this trading behavior realistic, even? Probably not.

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00:38:22.380 --> 00:38:33.249

Krzysztof Paruch: And the third question that, that I'm... haven't... that I haven't really, answered yet, or, like, haven't built much, support

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00:38:33.370 --> 00:38:37.969

Krzysztof Paruch: Around it... around is... How to...

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00:38:38.460 --> 00:38:48.069

Krzysztof Paruch: because now I have a very good understanding of the decentralized market structure and the logic behind it, but the micro-market structure of the order book

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00:38:48.330 --> 00:38:53.140

Krzysztof Paruch: is also something worth... worth investigating. So what I've been doing so far is

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00:38:53.310 --> 00:38:59.389

Krzysztof Paruch: kind of assuming, okay, there is kind of, like, you can set it up as an order book, or you can set it up as an AMM,

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00:39:00.000 --> 00:39:10.849

Krzysztof Paruch: Now, I've, and I've looked quite intensively into the derivation of how the AMM logic works, and the pricing, how it's, how it's developed there.

222

00:39:11.170 --> 00:39:30.390

Krzysztof Paruch: But there is a lot of literature that I've explored, around, order book microstructures, and how you can actually generate proper signals from the data that you are, looking at when you, when

you have order book information, and how this,

223

00:39:31.180 --> 00:39:40.629

Krzysztof Paruch: how you could build a unified framework around those things, because I don't think that what I've done here technically makes

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00:39:40.660 --> 00:39:51.820

Krzysztof Paruch: like, it's very robust. So I don't think that, like, by simply feeding in historical trades from an order book market into an LSMR,

225

00:39:52.310 --> 00:39:57.859

Krzysztof Paruch: is at all, representative. It's not a good comparison.

226

00:39:58.010 --> 00:40:08.890

Krzysztof Paruch: I think what you need to do is you have to investigate the microstructure of the order book, see how you can generate, like, a driving signal, and then

227

00:40:09.480 --> 00:40:15.600

Krzysztof Paruch: Develop maybe a framework that is unifying both approaches, and look at the performance of both.

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00:40:15.820 --> 00:40:22.740

Krzysztof Paruch: And... And then, you would have a very good, analysis, of,

229

00:40:23.210 --> 00:40:28.790

Krzysztof Paruch: In which case, you would be more leaning towards, kind of, like, one or the other.

230

00:40:30.080 --> 00:40:31.630

Krzysztof Paruch: implementation.

231

00:40:31.880 --> 00:40:36.060

Krzysztof Paruch: And... I don't know, that's this... Makes sense. This is, like...

232

00:40:36.360 --> 00:40:41.380

Krzysztof Paruch: Like, a good idea, or is this just, fishing in...

233

00:40:42.000 --> 00:40:44.099

Krzysztof Paruch: In a pond, that doesn't really make sense.

234

00:40:45.360 --> 00:40:56.300

Jamsheed Shorish: We have a couple of different metrics that you would be wanting to use to assess whether or not one or the other approach is one worthy of adoption, because you, on the one hand, you have sort of the overarching mantra about

235

00:40:56.420 --> 00:41:07.850

Jamsheed Shorish: there should be a metric such that I could tell whether a prediction market is well-functioning versus one that is not, on the one side. But then you have all of these other considerations, which is, you know, what's the market for?

236

00:41:08.150 --> 00:41:12.119

Jamsheed Shorish: And the market is, is, is not ultimately the market itself

237

00:41:12.260 --> 00:41:25.279

Jamsheed Shorish: to an individual trader who's posting in an order book, for example, they may not necessarily be concerned about whether or not the prediction market is good. They're actually interested in whether or not their prediction is right, because they actually get a payoff only under those conditions.

238

00:41:25.700 --> 00:41:43.340

Jamsheed Shorish: Right? So, there's going to be a series of criteria that you would be deciding, depending on what hat you're wearing. Are you the market maker? Are you the owner of the exchange? Are you the trader? Each of which is going to inform one or another type of submission process that's going to be beneficial.

239

00:41:43.390 --> 00:41:58.670

Jamsheed Shorish: And, of course, you can parameterize to maybe get different edge cases or different polar cases where you'd say, oh, I'd clearly want to have an AMM in this situation, I'd clearly want to have an order book in that situation that's centralized. There are decentralized order book implementations also out there that sort of sit in the middle.

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00:41:58.670 --> 00:42:07.090

Jamsheed Shorish: But you'd want to be looking at, kind of, the extremes first, before trying to nuance whether one or the other one is more appropriate.

241

00:42:07.270 --> 00:42:24.339

Jamsheed Shorish: So, I think maybe one of the overarching questions is, do you already have enough of your, like, understanding from the decentralization that you've already performed to really propose, or even to kind of stand behind metrics that you will be using in order to assess whether one or another is more appropriate?

242

00:42:26.830 --> 00:42:33.539

Krzysztof Paruch: Right, if you ask me this way, the answer is no, but, but this is also because I was...

243

00:42:34.430 --> 00:42:38.870

Krzysztof Paruch: like, so completely blindsided by the assumption that I really need

244

00:42:39.060 --> 00:42:47.270

Krzysztof Paruch: The, the logarithmic market to perform similar to the, to the, objective probability.

245

00:42:47.730 --> 00:42:51.460

Krzysztof Paruch: I have to open the door very quickly, let me just, be back in.

246

00:42:52.790 --> 00:42:53.420

Jamsheed Shorish: Sure.

247

00:42:57.580 --> 00:43:13.000

Jamsheed Shorish: maybe just interject while he's jumping out is just a... for those who haven't been following this series that Chris has been putting together, it's really been about the idea of building a kind of a means by which a decentralized prediction market can occur using some of the

248

00:43:13.000 --> 00:43:25.309

Jamsheed Shorish: The consistency foundations for price formation, and

in particular for utilizing scoring rules to say which ones are consistent and truthful, because you're looking for things that are giving you some truthful revelation about the objective

249

00:43:25.310 --> 00:43:39.830

Jamsheed Shorish: actual outcomes that may occur. And here, what's happening is that there's more and more... as more and more structure is being laid onto the implementation, some very interesting questions are being uncovered with regard to, you know, what is it that it's supposed to be doing, and how are individuals going to be interacting?

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00:43:39.880 --> 00:43:52.339

Jamsheed Shorish: in the same way, at least from my perspective, and it's a limited perspective, I grant you, but my perspective is that this is something which is very akin to how individuals interact in a betting framework.

251

00:43:52.340 --> 00:44:09.140

Jamsheed Shorish: In general, versus how the individuals who are actually the casino owners, or individuals who are promoting the betting terms, how they view the system. And those are two vastly different types of systems, and they have very different types of metrics to measure their, their relative performance, and even their participation and their structure.

252

00:44:09.160 --> 00:44:09.920

Jamsheed Shorish: So...

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00:44:10.460 --> 00:44:13.360

Krzysztof Paruch: Nice! Thank you so much for the overview, it's very nice to...

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00:44:13.360 --> 00:44:14.700

Jamsheed Shorish: Yeah, just, like, quick catch.

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00:44:14.700 --> 00:44:31.219

Krzysztof Paruch: step back and get it, yeah, very focused. But to answer your question, so, like, I'm obviously looking at it from the market designer perspective, so, like, market owner, market designer, because, like, I'm looking at it through the lens of, like, I want to build it.

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00:44:31.300 --> 00:44:43.960

Krzysztof Paruch: But, eventually, I mean, like, all the questions, all the perspectives that you mentioned are fair, right? Like, you can both, like, designing the market, look also at it through the lens of a participant and try to optimize for some behavior there.

257

00:44:44.410 --> 00:44:53.590

Krzysztof Paruch: Or you can just build some other stuff around that that would, allow you to reason, basically, from... through the eyes of market participants.

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00:44:53.770 --> 00:45:01.629

Krzysztof Paruch: So, yeah, clearly from the market designer's perspective, so the question would be, what is it that is appealing to a market designer?

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00:45:01.760 --> 00:45:11.829

Krzysztof Paruch: And what should you... what should you optimize for? And I, again, had the very naive understanding that what I want to optimize for is kind of like the...

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00:45:12.120 --> 00:45:15.190

Krzysztof Paruch: Information revelation, or, like, the probability

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00:45:15.600 --> 00:45:19.360

Krzysztof Paruch: interpretation of the data that I have collected.

262

00:45:19.680 --> 00:45:23.500

Krzysztof Paruch: Such that I can use it for some... something else. I can... I can just...

263

00:45:23.840 --> 00:45:38.710

Krzysztof Paruch: I don't know yet why or how. I can use it for... just to inform something else, right? So I've collected information, I have estimated probabilities, market probabilities of future events, and I can make something out of it.

264

00:45:39.990 --> 00:45:44.519

Krzysztof Paruch: But maybe it's worth revisiting this.

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00:45:45.490 --> 00:45:54.109

Krzysztof Paruch: Okay, so, so this is, kind of like, the experimental arc here. I mean, we can obviously look in, in all of the

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00:45:54.540 --> 00:46:15.069

Krzysztof Paruch: data, how it looks like, what you're able to pull, how order books look like, how prices look like. And I'm reasoning right now about how to generate signals that I can then feed in into those markets and, like, make a proper comparison, and then define metrics that would help me to understand those differences. I think that this is a very neat result that you can publish.

267

00:46:15.590 --> 00:46:25.659

Krzysztof Paruch: And then you can just, like, make some claims about, like, performance of markets or marketplaces under different, settings.

268

00:46:25.780 --> 00:46:35.370

Krzysztof Paruch: But, I'm also doing other things, which is, for example, looking at...

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00:46:36.480 --> 00:46:48.629

Krzysztof Paruch: at this publication. So this is, like, a directly, like, a direct paper that is, comparing market scoring rules and continuous double auctions, or price discovery.

270

00:46:49.310 --> 00:47:00.299

Krzysztof Paruch: And this is one that has been just recently published, and one of the few I found, I mean, I will be digging deeper into kind of, like, this arc of literature.

271

00:47:00.570 --> 00:47:13.050

Krzysztof Paruch: But, on a very high level, what's happening here is, like, I haven't studied it very well, like, I just skimmed through it. But, what they say here is that, both

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00:47:13.260 --> 00:47:18.740

Krzysztof Paruch: Cases have their... Benefits.

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00:47:18.790 --> 00:47:34.360

Krzysztof Paruch: And you should use, the market scoring rules for illiquid markets and the continuous double auctions for liquid markets. So, the argument that they make here is, okay, you don't have

to subsidize,

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00:47:34.360 --> 00:47:39.249

Krzysztof Paruch: or the book market, you do have to subsidize a decentralized market.

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00:47:39.900 --> 00:47:42.780

Krzysztof Paruch: And, if you want to

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00:47:42.930 --> 00:47:47.639

Krzysztof Paruch: Allow for trading. So, like, if you want to just even allow people to step in.

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00:47:47.820 --> 00:47:59.719

Krzysztof Paruch: You should be using, the subsidy, and you should be using the market scoring rule, but for kind of, like, natural flow of information, when you have, like, bigger communities, bigger trading activity, it's completely sufficient.

278

00:48:00.460 --> 00:48:20.239

Krzysztof Paruch: to do, the continuous double auctions, or the centralized order bookstyle solution. And what they do here, it's not kind of like the signal processing perspective that I'm looking at. What they're doing here is some form of an analysis where they try to get,

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00:48:20.990 --> 00:48:23.860

Krzysztof Paruch: Pri- the price discovery share?

280

00:48:24.220 --> 00:48:29.540

Krzysztof Paruch: of different markets. So, like, they tried to compare how markets

281

00:48:29.890 --> 00:48:38.020

Krzysztof Paruch: How informative they are with respect to revealing true prices, if you have both

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00:48:39.290 --> 00:48:43.470

Krzysztof Paruch: Markets available at the same time, and you allow people to trade against them.

283

00:48:44.130 --> 00:48:50.519

Krzysztof Paruch: This is what I know so far, but it's basically kind of like the starting point of this

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00:48:50.760 --> 00:48:59.070

Krzysztof Paruch: branch of analysis that will hopefully allow me to find some conclusions, that I can map to the

285

00:48:59.480 --> 00:49:03.160

Krzysztof Paruch: signal processing arc that I'm implementing here.

286

00:49:04.140 --> 00:49:23.270

Krzysztof Paruch: Yeah, I'm sorry, I cannot tell you more. I really didn't read the paper yet. But I found it very, very nice that something like this exists, and it got me intrigued into looking more into this direction, obviously, to find something similar, or something that allows me to make proper analysis.

287

00:49:24.430 --> 00:49:29.359

Krzysztof Paruch: Of doubles, of, of both setups.

288

00:49:32.090 --> 00:49:39.210

Krzysztof Paruch: Does it make sense to talk about this, or is it just, like, to... like, even if I don't know anything about it, probably doesn't make much sense, right?

289

00:49:52.800 --> 00:49:57.389

Zargham: Yeah, it seems like it would be better to come back to after people have a chance to read it.

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00:49:58.280 --> 00:50:03.359

Krzysztof Paruch: Yeah, yeah, no, sure. I mean, it's, like... Yeah, I agree.

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00:50:03.650 --> 00:50:12.849

Krzysztof Paruch: So this is what I have, or what I wanted to show you, at least today. I mean, we can dig deeper into some of the things, but...

292

00:50:14.030 --> 00:50:29.090

Krzysztof Paruch: Yeah, trying to build intuition around micro-market structures in both cases, trying to find a unifying framework on how to analyze their performance and the signal processing capabilities.

293

00:50:29.490 --> 00:50:42.229

Krzysztof Paruch: and make some interpretation about, performance, or price discovery, or information aggregation. So, there are some starting points here and there, but generally, I think that this is

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00:50:43.700 --> 00:50:48.919

Krzysztof Paruch: the question I have to answer in order to be able to make, make a good assessment of

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00:50:49.570 --> 00:50:56.450

Krzysztof Paruch: Which market makes sense for which use case, and how you should build it, and how you should utilize it.

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00:50:58.260 --> 00:51:12.359

Jamsheed Shorish: One of the nice things is that you've built a lot of tools that I think are really useful to propose that research track that you originally have been progressing on, which is the idea of being able to use one realization, or even better, a set of realizations.

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00:51:12.360 --> 00:51:19.179

Jamsheed Shorish: Of a prediction market's outcome in future prediction, or as a refinement of a model, or as, you know, a

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00:51:19.210 --> 00:51:38.259

Jamsheed Shorish: to be mercenary, you know, a packaging of a particular set of probabilities that you have, you know, a comparative advantage over for someone else to license, and so on. So one of the things that might be... this might be a good time to try, and I'm not saying it's the direction you have to go, depending on the directions you would like to go, but since you have the tools that you've been building.

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00:51:38.350 --> 00:51:46.680

Jamsheed Shorish: Might be to go into a very well-defined, not very complicated, but a well-defined, simple, synthetic.

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00:51:46.820 --> 00:51:57.689

Jamsheed Shorish: set of Monte Carlo realizations. So basically run a lot of simulations with the same initial conditions that are structured to provide different realizations.

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00:51:57.850 --> 00:52:16.010

Jamsheed Shorish: Different realizations of trades, different realizations of arrivals, different realizations perhaps within some boundaries of objective probabilities, if you want to keep those there, or if you want gradual information revelation to occur over time on the basis of trades and trading depth and so on. Those are all free parameters you can choose in the simulations.

302

00:52:16.100 --> 00:52:23.330

Jamsheed Shorish: And then try, and this would be kind of the overarching thesis, to figure out the function...

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00:52:23.450 --> 00:52:28.560

Jamsheed Shorish: That maps the actual, unknown, true, objective probabilities?

304

00:52:28.740 --> 00:52:31.149

Jamsheed Shorish: To the observed market outcomes.

305

00:52:31.420 --> 00:52:50.289

Jamsheed Shorish: And that would be some kind of a data-fitting exercise, or some kind of a very, like, nonlinear, naturally, in this case, I think, essentially a regression analysis, where you have this massive corpus of synthetic data to be able to perform that in an effective fashion, whether that's machine learning, or whether that's, you know, it's a canonical regression.

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00:52:50.290 --> 00:53:04.569

Jamsheed Shorish: you would be able then to say, aha, here is the way in which all these things could happen, and each time I run one market realization, I rerun the universe with a different set of initial conditions, that's giving me incrementally more information about that mapping.

307

00:53:04.570 --> 00:53:18.450

Jamsheed Shorish: to at least get a feeling for, okay, since that's kind of the multiple universes realization of this functional form, are there invariant properties that I can extract that would be occurring

308

00:53:18.500 --> 00:53:23.710

Jamsheed Shorish: even if I ran one realization for one specific thing one time.

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00:53:24.030 --> 00:53:31.899

Jamsheed Shorish: Or a small number of a class of things that are related, like elections, or like Super Bowl victories, or, you know.

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00:53:31.960 --> 00:53:33.800

Jamsheed Shorish: Football outcomes in general.

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00:53:33.860 --> 00:53:47.969

Jamsheed Shorish: That would at least be able to provide kind of an ability to both leverage the tools you've already used, and also kind of move forward in your understanding of how to interpret market outcomes vis-a-vis the actual objective probabilities that are being

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00:53:47.970 --> 00:53:54.210

Jamsheed Shorish: essentially signal process. They're being processed by the aggregation system of the trades to get a realization.

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00:53:55.610 --> 00:54:00.500

Jamsheed Shorish: So, it's just a possible path forward, would be this at this point. Seems natural.

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00:54:02.700 --> 00:54:08.660

Krzysztof Paruch: Okay, I'll take this up, as a potential way.

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00:54:13.440 --> 00:54:19.420

Krzysztof Paruch: Okay, so running Monte Carlo simulations to... investigate...

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00:54:20.660 --> 00:54:25.360

Krzysztof Paruch: if there are some recurring patterns or recurring themes, and I could derive.

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00:54:26.940 --> 00:54:43.379

Jamsheed Shorish: Well, two things, right? First, you would be... you'd be cheating in the sense that you have all of these different

universes that don't really exist. You'd be creating them synthetically in order to be able to figure out the mapping between objective probabilities and actual market outcomes. Once you've figured that out.

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00:54:43.430 --> 00:54:55.290

Jamsheed Shorish: Then there are properties of that function, which may look completely unknown in the real world, because you only have a limited data set, or almost, like, one point, which you can't do a regression at all on, but there may be properties

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00:54:55.420 --> 00:55:00.599

Jamsheed Shorish: Of that discovered mapping, That themselves are invariant.

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00:55:01.100 --> 00:55:11.350

Jamsheed Shorish: And so you're discovering the kind of the way that the aggregation process is performing, its function has certain things that, on the basis of that first step.

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00:55:11.590 --> 00:55:16.910

Jamsheed Shorish: Are enabling you, then, to make the claim, okay, this is actually representative of a particular class.

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00:55:16.910 --> 00:55:18.169

Zargham: In a, A prediction.

323

00:55:18.200 --> 00:55:38.140

Zargham: In a simulated universe in the worlds that Jumpstreet's describing, you can create scenarios where you actually know the probabilities a priori, and the market mechanisms get played out against, you know, whatever the scenarios you create are, and so you can ultimately use it as an experimental bed for modeling

324

00:55:38.140 --> 00:55:54.850

Zargham: the error, basically, modeling the relationship between the underlying probabilities and the resultant trajectories in the prediction markets. At least, if you think of it, going back to our previous work on markets and signal processing, we don't look at circuits and go.

325

00:55:54.850 --> 00:56:19.679

Zargham: they always estimate everything perfectly! That would be ridiculous. What we do is that we build engineering simulations in order to understand, in addition to mathematical models, in order to understand the steady-state error, as well as the various ways in which our design choices affect trade-offs between, let's say, latency in the estimates versus, like, biases in them, etc.

326

00:56:19.680 --> 00:56:42.579

Zargham: and in this case, I think you have enough machinery to effectively, if you continue thinking about this as a signal processing problem, then there's some unknown in the real world signal, but in the simulations, in these alternate universes, you can... you can know the underlying statistics, because they're the statistics of the generator processes in your simulations, so then you make lots of experiments.

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00:56:42.580 --> 00:56:49.009

Zargham: And you collect data about the ways in which the market structure does or does not,

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00:56:49.010 --> 00:57:13.879

Zargham: track that, and then you build, basically, yeah, a model... I think of it as building a model of the error. And then you could potentially exploit any structure you find. So if you learn that there's structure in the error, now you've identified something that you could leverage to create something that is... I won't necessarily say more efficient, because to your point earlier, efficiency may not actually be the metric that you as the market designer are optimizing for, but you could... you can intel... you know.

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00:57:13.880 --> 00:57:32.300

Zargham: intelligently design market structure, even in... with multiple competing metrics that you are kind of tailoring against, but once you have that kind of model of the error, you have something that would allow you to find a new point in the Pareto optimal space and the market design... in the market design space.

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00:57:32.910 --> 00:57:37.349

Zargham: Is that consistent with what you're saying? Because that's kind of how I interpret it.

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00:57:37.350 --> 00:57:49.809

Jamsheed Shorish: Yeah, exactly. Yeah, and that's, like, even more fine-grained in the sense that you're actually able to build out that

notion of error, which I think would be quite important once you've actually figured out, like, how are you mapping one to the other, and that's, again.

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00:57:49.810 --> 00:58:00.709

Jamsheed Shorish: like, it fits, from my perspective, naturally to, Chris, what you've already created in terms of your forays into signal processing, because that's something that's basically a signal processing exercise. Yeah, definitely.

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00:58:03.030 --> 00:58:12.130

Krzysztof Paruch: Nice. I mean, is it, like, am I thinking about this correctly if I say, okay, I'm gonna build experiments or Monte Carlo simulations around, like,

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00:58:12.440 --> 00:58:31.110

Krzysztof Paruch: coin flip, so where I know that, kind of, like, this experiment is, like, a 50-50 probability, and then I can, like, define, like, some experiments around that, and then I would know what, kind of, like, the outcome should be, and then I can compare the simulation runs against this. Is it kind of, like, the right direction to think about it?

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00:58:32.000 --> 00:58:37.100

Zargham: I think that would work as a starting point, but the other thing I would keep in mind is that

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00:58:37.100 --> 00:58:52.589

Zargham: when you move to signal processing, you don't have to assume that, the, the value is a static value. It may not be, like, the probability of a coin flip. It might be that you want something that represents,

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00:58:53.210 --> 00:59:17.279

Zargham: Let's say, like, given there's lots of other things going on in the world, at any given point in time, there may be a probability, and that itself might be drifting up and down as a result of external factors, and you could imagine modeling them away, but if you have, essentially a sequence of probabilities over your time, it has to collapse at the end, so when it's resolved, the probability has to become, you know, 1 or 0.

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00:59:17.280 --> 00:59:41.630

Zargham: But during the transient, up until the point that it collapses, you could still model that signal out, and you could still generate trajectories where that is drifting, and that there's partial observability or some incomplete information that the market is reacting to. It doesn't know the probability, but it knows it may be getting some derivative information from which the, from which the

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00:59:41.830 --> 00:59:50.700

Zargham: actor-level traits are occurring. And so you could end up with a slightly more complex block diagram, where there is a ground truth signal, which is unknown.

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00:59:50.820 --> 00:59:55.889

Zargham: it is itself drifting, right? Again, we kind of did this with our,

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00:59:55.890 --> 01:00:13.509

Zargham: Economic Games as Estimators paper. I'd have to go back and look at exactly what we did, but it... I'm describing something that's closer to that family, where there's... there's... there's the difference here being that the... it's a probability of an event, and when the market closes, it has to have collapsed to 0 or 1.

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01:00:13.510 --> 01:00:19.070

Zargham: As opposed to it being a free-floating price that continue... is in continuity.

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01:00:19.800 --> 01:00:25.569

Zargham: But it could still be changing over time, because in the real world, you know, I would argue that

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01:00:25.570 --> 01:00:42.729

Zargham: it isn't known, so it could just be that there's a... the representation is the current probability is, and that value drifts over the time period, and that's the signal that this thing is ostensibly trying to track, right? The job of the prediction market is to try to track

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01:00:42.760 --> 01:01:05.410

Zargham: that probability, and the incentives are derived from the closure point. The collapse down to 0 or 1 is the point at which you resolve incentives and payouts, but during the intermediate period, we don't assume that there's a known true or false, or even a known fixed

coin flip probability, but rather, you know, an evolving set of odds.

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01:01:05.410 --> 01:01:09.900

Zargham: As a result of information that might unfold in the intervening time.

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01:01:11.030 --> 01:01:15.610

Zargham: But I would still probably model that as a signal of

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01:01:16.270 --> 01:01:27.830

Zargham: probability value, so it's 0, 1 at each point in time, and then your discrete time chunking will affect your model, anyway. I mean, does that make sense? Is that track for you?

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01:01:27.830 --> 01:01:46.940

Krzysztof Paruch: Yeah, no, it makes perfect sense. I mean, I'm just, thinking how... where to position this, kind of like in the research arc, but because, like, I'm at a stage where I want to wrap this up a little bit, to some extent, like, to get some conclusions, or, like, some information out of it, and then move on with the next step, but, it sounds definitely, like, like the next.

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01:01:46.940 --> 01:02:05.919

Zargham: If you're trying to make a model of the error, it's a function of functions, right? Like, you have the function... one function is the process implemented by the market itself, which produces an estimate at any given time, and the other is a,

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01:02:06.510 --> 01:02:31.400

Zargham: a hidden one, which is the function that produces the current probability at any given time. And so, when you're building your simulation, you're saying, like, hey, there's the generator function that produces the sequences of probabilities which necessarily collapses to 0 or 1 at the end, and the other is a generative process that emerges from interaction with the market, which produces a sequence of probabilities between 0 and 1, and then collapses

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01:02:31.400 --> 01:02:34.929

Zargham: at the end. And so the question is those two signals.

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01:02:34.930 --> 01:02:58.160

Zargham: how are you, you measuring... how are you comparing them? And

so if you can find a model of the relationship between those two signals for a given, for given assumptions about the underlying market process and the market structure, then I think that's your... where your conclusions would come from. They would come from statements about any structure that you discover.

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01:02:58.160 --> 01:03:02.209

Zargham: In the, in the relationship between those two, signals.

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01:03:02.570 --> 01:03:09.880

Zargham: Again, drum sheet, spot check, I'm... I came in a little late, so I want to make sure I'm not, like, jumping tracks.

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01:03:10.250 --> 01:03:27.519

Jamsheed Shorish: No, no, that's exactly it. And yeah, it is kind of like, Chris, from your perspective, it just... it kind of depends upon where do you want to stick a pin in this and know that you need to move forward, but it would be good probably just in... even in the case that you didn't want to proceed in that direction, just to write that down and write it up, and kind of say, okay, this is a thing I can do.

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01:03:27.520 --> 01:03:37.449

Jamsheed Shorish: And, like, create a folder of, these are the kinds of things that I can do now that I've built the machinery, because it would kind of be a shame not to be able to exploit that to its fullest potential, as Zardim has mentioned.

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01:03:38.280 --> 01:03:38.890

Krzysztof Paruch: North.

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01:03:40.660 --> 01:03:41.520

Krzysztof Paruch: Cool.

360

01:03:42.380 --> 01:03:48.029

Krzysztof Paruch: Yeah, appreciate your feedback, thank you so much for listening and tuning in today.

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01:03:49.280 --> 01:03:54.850

Krzysztof Paruch: Yeah, it's very cool to be able to check those ideas against kind of like, this group of people.

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01:03:55.130 --> 01:04:11.870

Krzysztof Paruch: Yeah, so I've, optimistically set, the next meeting for in 28 days, I guess, which, like, we will see where we will get by then, but, so far, I very much like the pace and, like, the way how we move forward with that, so looking forward to the next conversations.

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01:04:12.580 --> 01:04:26.280

Jamsheed Shorish: Outstanding. Yeah, no, appreciate the time, the continued amount of input that you give us. It's always great to be able to track it, and I'm hoping, of course, it's useful on your side, but it's definitely great to see how you're pushing the envelope in different directions each time you're coming around again, so thanks.

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01:04:26.870 --> 01:04:30.320

Krzysztof Paruch: Sounds Thank you very much, guys. See you soon.

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01:04:30.600 --> 01:04:31.240

David Sisson: Thank you.

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01:04:31.240 --> 01:04:32.420

Jamsheed Shorish: Thanks, everybody. Thanks, Chris.

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01:04:32.590 --> 01:04:33.370

Jamsheed Shorish: See you.